A concurrent, minimalist model for an embodied nervous system

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***Abstract*—The nervous system has inspired many computational simulations modeling systems capable of learning through interactions with their environment. We propose a minimal model of a nervous system operating in a concurrent manner, capable of maintaining behavior homeostasis of a virtual organism. The model nervous system includes subsystems which, according to the functions performed, may be classified as sensory, effector, cognitive or emotional. These subsystems, while autonomous, exchange messages between one another. The organism is endowed with innate behaviors of exploration, approach, eating and resting; the nervous system affords the possibility of learning through evaluation and memory of experiences associated with these behaviors, interacting with a virtual environment. These experiences influence the selection of subsequent actions, to promote self-regulation, and thereby, survival of the organism. The proposed model is concurrent, asynchronous and non-deterministic. We simulate a simple realization of the model. Our results show that the expected behaviors of establishing food preferences in fact emerge. We conclude that the model is functional and robust and that learning increases longevity.**

I. Introduction

The development of a nervous system depends on the actions and history of the organism in its environment. From this perspective, thoughts, concepts and perceptual categorizations are based on interactions between brain, body and the environment ([1]). The evolutionary developmental approach to cognition begins with a situated vision of organisms in their environment, as explained by various authors ([2], [3], [4], [5], [6], [7]), who are in fundamental agreement regarding the principle that organism and environment are inseparable, possessing a reciprocal relation, and must be understood as a co-evolving whole.

These ideas have influenced research in artificial intelligence (AI) since the early 1990s ([8], [9]); according to [7], the notion of embodied cognition is essential for modeling intelligent systems. Various bio-inspired models of the brain have been elaborated ([10], [11], [12], [13], [14]) as well as biologically inspired cognitive architectures (for a review see [15], [16] and references cited therein). Interesting models in this context have also been developed as entertainment software agents ([17],[18]).

Florian [13] asserted, 'genuine, creative artificial intelligence can emerge only in embodied agents capable of cognitive development and learning by interacting with their environment'. This author simulated a model including an agent with innate behaviors, capable of learning through its experiences, and highlighted the importance of drives or reflexes to impel the agent to explore its environment. Without such exploration, learning would not occur.

Fujita [19] sought a realization of 'intelligence' as opposed to symbolic manipulation, the latter being the hallmark of conventional AI. By proposing 'Intelligence Dynamics,' he tried to integrate the ideas of embodiment and dynamics in a constructive approach. Fujita [19] also incorporated intrinsic motivations in his models, which permit the continual functioning of control, planning, prediction and action. In the two models proposed, which are organized in hierarchical layers, [19] is concerned with furnishing the conditions for spontaneous learning.

Agmon & Beer [20] adopted a perspective he called 'embodied dynamicism', 'which sheds the assumptions of higher-level decision-making process and instead commits only to dynamical systems explanations. This allows for more integrated architectures, which embrace nonlinearity and aim to describe behavior as emerging from rich interactions across the brain, body, and environment.' This author proposed artificial life models with autonomous agents which select actions guided by their internal state as well as sensory input.

Ignasi Cos & Hayes [14] devised a model organism that learns to adjust its consumption in an adaptive manner. The agent possesses drives characterizing the urgency of action based on bodily needs, which motivate its actions. Despite the emphasis placed on the dynamics of sensory-motor interactions and homeostasis, which typify the process studied, the model is sequential in nature.

Our objective in the present work is to create a functional minimal model of a nervous system, understood as an embodied system, which possesses subsystems operating concurrently, in an asynchronous and nondeterministic manner. Based on the operation of this dynamical system, the global behavior that arises should be coherent, and capable of maintaining behavior homeostasis of the virtual organism. This organism is endowed with innate behaviors of exploration, approach, eating and resting. The bodily needs or drives of hunger and need for rest must be satisfied if the organism is to survive.

We emphasize the concurrent, asynchronous, nondeterministic operation of subsystems associated with visual and tactile perception, eating, and movement, within the model nervous system. Nevertheless, given the extreme complexity of real nervous systems, we allow ourselves to include constructs and characteristics not directly related to biological reality. Thus, in the interest of simplification, the model includes entities which, while performing a biological function, do not correspond to a unique anatomical structure. Specifically, we employ elements such as 'HomeostaticRegulation', 'Valuation', 'Partial Appraisal', 'Full Appraisal' and 'Locomotion Effector' which subsume diverse cerebral regions and interactions.

II. Theoretical and Conceptual Issues

A. Why emotion in a cognitive model?

Phylogenetically older cerebral structures, such as the 'limbic system', formerly understood as specifically responsible for the manifestation of emotions, highlight the fact that emotions are necessary for survival, since they enable animals to explore, communicate and interact more effectively with their environment ([21]). Despite the importance of the 'limbic structures' [1] from both the historical viewpoint and that of emotional expression, emotions cannot be restricted to a specific region of the brain ([22]).

Similarly, advances in neuroscience indicate that complex feedback loops connect distant brain regions, without any explicit subordination. Such evidence suggests that the concept of a sequential top-down hierarchy requires revision; such a hierarchy is not found in the prosencephalon ([23]).

From the neuroanatomical point of view, as well as that of brain dynamics, there seems to be little reason for separating emotion from cognition. Thus we consider it essential to include emotion, in the restricted sense explained below, in our model.

B. What is emotion?

Buck [24] proposes a classification of affects in which phylogenetically older motivational systems are involved in the manifestation of innate (non-learned) reactions. These systems are, however, also involved in more complex emotions. In order of increasing complexity, [24] classifies motivational/emotional systems as: reflexes, instincts, bodily needs and affects. He further proposes that emotional manifestations are perceived in distinct levels: emotion I (arousal)[2] important for adaptation and homeostasis; emotion II (expression) involved in social coordination; and emotion III (experience) which involves self-regulation through subjective experience, sentiments, desires and affects. In the present work we consider emotion in the form of bodily need, restricted to the level denoted emotion I in the scheme of Buck [24]. Inspired in Hebb [26], behavioral efficiency E was modeled (at least for simple tasks) to the level of emotional arousal A through a concave, quadratic function

$$E = -aA^2 + bA + c \tag{1}$$

such that there is an optimal level of physiological excitation, at which efficiency is maximum. In the present work we use the parameter values $a = \frac{-40}{49}$; $b = \frac{40}{7}$; $c = 0$.

In the model, behavioral efficiency governs the speed of movement. In this sense, the level of arousal (emotion I) affects the ability of the organism to navigate its environment.

C. What about homeostasis?

The notion of homeostasis assumes a central role in the incorporated approach to cognition. Homeostasis is broadly understood as a state of dynamic equilibrium, attained when all the internal systems of an organism work in harmony to maintain its stability. Our understanding of behavioral homeostasis is similar to the idea proposed by Walter Cannon ([27], [28]), who sees organisms as open systems, responsive to changes in the environment, and subject to fluctuations as it varies. The resulting fluctuations in physiological variables are nevertheless maintained within narrow limits by internal regulation mechanisms. Reviews on the concept of homeostasis can be found in ([28]) and ([23]).

[1] Discussion continues regarding the definition of these structures and, for this reason [22] suggest that the notion of the limbic system should be abandoned.

[2] One of the most important and fruitful aspects of the arousal theory was that it assumed that arousal can be measured directly, by psychophysiological measures that reflect autonomic, somatic, and central nervous system activity ([25]).

In our model, the internal subsystems designated as Partial and Full Appraisal, and Homeostatic Regulation, function to maintain the dynamic equilibrium of the organism, based on its physiological necessities, sensory-motor capacities and cognitive functions. These subsystems confer a measure of autonomy to the organism, in the sense that it responds to environmental stimuli with the aim of satisfying preprogrammed objectives (i.e., satisfying its needs for food and rest), with external intervention, in an environment that is unknown a priori.

D. Affordances

In the situated approach, the cognitive-emotional process is seen as an indivisible whole: one cannot be understood without the other. Cognition and emotion are co-dependent ([29], [30]). But how does this process occur?

Following the suggestions of Gibson [3] and Maturana & Varela [5], we see this as the organism actively identifying the relevant features in its world. In this process, called *affordances*, the organism constructs distinctions, thereby identifying features relevant for action.

III. MODEL OF THE ORGANISM-ENVIRONMENT SYSTEM

We regard the nervous system as an open dynamic system ([31]), and living beings as open systems ([28]). Our model treats the brain and body as inseparable, and possesses an internal dynamics that is concurrent, asynchronous and non-deterministic. The dynamics of the model is implemented via multiple threads. [3].

Much of the complexity of the model arises from its biologically inspired nature. An example is the apparent duplication of elements, e.g., the sensory organ 'eye' and the internal sensory system 'vision sensor.' At first glance one might suppose that one of these might be eliminated. The structures represent, however, an external organ and an element within the nervous system, respectively, which operate via mutual interactions. Thus the threads internal to the virtual organism are inspired by neurofunctional aspects of biological organisms, while realizing a simple, local task. That is, each thread simply follows a cycle of receiving messages, changing its internal state, and emitting messages.

Figure 1 shows the internal subsystems of the virtual organism, which bear names alluding to their functions. In the upper part of the diagram are the sensory organs, which coincide in large measure with the motor organs appearing in the lower part, except for the 'skin', which does not assume a sensory function in the present work.

Except for the sensory-effector organs[4], represented in Figure 1 by rectangles, the remaining subsystems comprise the nervous system. All the structures enclosed in dotted lines represent threads executing independently, which exchange messages amongst one another. The remaining elements, enclosed by solid lines, do not operate as threads, but cooperate with the latter. The sensory system mediates the messages received from the sensory organs. In addition to this subsystem, the nervous system comprises the emotional, cognitive, memory and sensory subsystems, whose operation is explained in the following section.

As mentioned above, behavioral efficiency is assumed to be a concave function, see 1, of the level of deprivation ([26], [25]). Thus, as the level of deprivation of bodily needs increases, particularly in the case of hunger, the efficiency in satisfying it grows until an optimal level is attained; at this point, speed of movement take its maximum value; beyond this point, the speed decreases.(See Figure 8)

Another important aspect of the model is the modulation of sensory capacities, through variation of the field of view, which becomes smaller as the organism approaches an object. This highlights the object selected and reduces the likelihood of new sightings. The opposite occurs when there is no object in the visual field (see Figure 2). These changes in effect permit the organism to scan the environment, as biological organisms do, typically through movement.

A. Cognitive-emotional process

The cognitive-emotional process occurs via interactions with the environment and via internal messages. The sensory organs receive stimuli from the environmental, while motor organs act within or upon it. Environmental stimuli are of three kinds: nutritional, visual and mechanical; each is perceived by a specific sensory organ. Similarly, each type of internal message is received by a specific subsystem or subsystems.

All the structures that operate via threads realize an operation consisting in (1) receiving a stimulus or message; (2) processing it; (3) generating new messages, which are perceived by other structures.[5] Messages are exchanged via buffers shared amongst the threads. These buffers contain all the conditional locks which characterize the relations between threads emitting messages and those receiving messages. The conditionals lock certain threads from executing until the most recent message emitted by it has been received. For other threads, execution is not locked by the buffer. These locking conditions are necessary because the order of execution of the threads is not fixed, or controlled by any supervising agent.

For purposes of illustration, we describe two examples in which the organism interacts with its environment; the first involves eating a fruit, followed shortly thereafter by an evaluation of this experience. Suppose that initially, the organism is touching (and seeing) one or more fruits. In this case, eating one counts among the set of possible actions. The internal operation in this case may be described via the steps enumerated below, remembering that the order of execution of the threads might not follow the sequence described. A possible sequence of events, then, is as follows:

[3] In computation, a thread is a unit of execution, i.e., a component of the program that can be executed in a concurrent manner with other threads. Multiple-thread programming therefore permits one to simulate parallelism, since a single processor realizes 'simultaneous' execution of all the threads designed to run concurrently.

[4] The arrows leaving the lower part of Figure 2 represent interactions between the body and the environment. These interactions occur via stimuli designated as: mechanical, luminous and destructive, the latter corresponding to removal of fruit when 'swallowed' by the mouth organ.

[5] Certain threads may, in particular cases, not generate a new message.

1) Having neared the fruit, the organism can turn and touch it, in which case the Eye receives a visual stimulus ('image') of it, while, at the same time, the Mouth receives a mechanical stimulus corresponding to the fruit. (Recall that, for simplicity, we assume that the Mouth is the organ that receives mechanical stimuli associated with eating. If another part of the body receives such a stimulus, the affordance of eating does not arise.)
2) Having received a visual stimulus corresponding to a fruit, the Eye emits a visual message, which is recognized by the nervous system.
3) Concurrently, the Mouth emits a tactile message, recognizable by the nervous system.
4) The Vision Sensor receives the visual message and sends a Spatial message to the buffer called InnerMessagePool.
5) The Spatial message is received by Partial Appraisal, which begins its operation. The Spatial message communicates to Partial Appraisal part of the current internal state of the organism, together with the environmental situation (what is visible). In addition, complementing the information about the environmental situation, a Tactile message, signaling that something has been touched, is received by Partial Appraisal. The internal state is also characterized by the levels of deprivation, and drives. On this basis Partial Appraisal elects the drive exhibiting the highest level of deprivation as the one to be to be attended, and adjusts the Behavioral Efficiency (that is, the length of the next step taken by the organism). This partial appraisal is then grouped into an Emotional message generated to Full Appraisal. At the same time, Partial Appraisal produces an Increase message, responsible for maintaining the metabolic activity, to stimulate HR, corresponding to a correction in the levels of deprivation.
6) Stimulated by the Increase message, HR updates the deprivation levels of the drives Hunger and Rest, (that is, the level of emotional Arousal), in this case, increasing the levels of both. [6] With this, HR terminates its activity. Up to this moment, the organism, although touching the fruit, has not eaten it; we assume that the organism has not chosen to rest.
7) Meanwhile, the Affordances[7] are established, according to the situation. In the case under discussion, the organism can eat the fruit, avoid it, rest, or, if there are other fruits in its field of vision, choose to approach one of the latter.
8) Concurrently, an Emotional message is received by Full Appraisal (FA), which performs a full evaluation of the internal state of the organism, as well as the current state of the environment, so as to select the best action available. To this end, FA activates Action Selection (AS).
9) If the selection criteria used previously have not led to a unique choice of action (see Algorithm 1), AS performs a search in Long Term Memory (LTM), for past experiences of the same kind, and their emotional value, to assist in the decision of which action to perform now.
10) Once the action to be performed has been selected - say, eating the fruit with which the organism is in contact - AS alerts FA, which registers this action in Working Memory as the next to be executed. (In the near future this action will receive an emotional evaluation, to be registered in LTM.) Next, FA generates a Cortical message alerting the effector system to realize the recently selected action.
11) The Cortical message is received by three effector components: the Mouth Effector, Focus Effector (FE)[8], and Locomotion Effector (LE) [9].
12) The Mouth Effector, on receiving the Cortical message, emits a Somatic message for the Mouth to eat the fruit with which it is in contact.
13) The Somatic message is received by the Mouth, which eats the fruit. The latter is removed from the environment.

Once an action is realized, it is evaluated. In the present case, the evaluation proceeds as follows:

1) The organism receives a Nutritive message from the Mouth, which is transformed into a Nutritional message, signaling that something with nutritional value was eaten;
2) On receiving the Nutritional message, the Gustatory Sensor emits a Decrease message;
3) Having received the Decrease message, HR updates the level of Deprivation of the drive Hunger, reducing it in this case, according to the nutritional value of the fruit eaten. HR then emits a Valuation (V) message;
4) The V message is received by Valuation, which verifies the value (positive or negative) of the recent action. If there is no record of this kind of fruit in LTM, such a record is created, including its emotional value. In this manner, this experience evaluated emotionally, enters the catalog of experiences, which may be consulted by FA to aid in the selection of future actions. If such a record already exists in LTM, its emotional value is updated.

The weight W_i of an experience in LTM is updated as follows,

$$W_i = W_{i-1} + Q_i \qquad \forall i \in \{fruits\} \tag{2}$$

Where Q is the nutritional value of the recently consumed fruit and i is the index of the fruit eaten. [10] The emotional value of

[6] This increment is fixed, given by the Δ in Table II.

[7] See TableI for a list of possible affordances.

[8] This effector focusses attention on the object in view. In this case, however, the object is touching the organism, so that such an attentional focus is unnecessary. Thus the FE is not activated in the present example.

[9] The LE signals the Body of the need to move in order to realize the action selected. In the present case no such movement is required, and so the LE is not activated.

[10] The weight of the action *resting*, while not considered in the present example, is updated in an analogous manner.

TABLE I
AFFORDANCES: POSSIBLE SITUATIONS AND ACTIONS.

Target	Situation		
	Seeing	Seeing and touching	Not seeing
Redfruit Greenfruit Grayfruit	to approach to avoid to rest	to eat to avoid	to rest to wander

an experience stored in LTM takes but two values, *true* or *false*. Let D be the deprivation level of the emotion attended and N be nutritional value of fruit consumed. Then the emotional value is *false* if $N + D < 0$ and *true* otherwise. Note that the two interactions described above are not the only ones the organism is capable of. All of the affordances described in the following section represent possible interactions. In addition, we note that at any moment during the chain of events characterizing an interaction, another stimulus or message may arise, provoking its own sequence of events, independent of (and not synchronized with) the first (see Figure 1) . It is also possible for two or more chains to converge on a given internal subsystem, which may then emit a single message, resulting in a single behavior.

B. Affordances

The structure Affordances in Table 1 corresponds to the combinations of situations the organism may face and its respective actions. For example, if the organism sees a target of the kind Redfruit, Greenfruit ou Grayfruit, the possible actions are ‘to approach’, ‘to avoid’ or ‘to rest’. If it both sees and touches a fruit, the possible actions are ‘to eat’ or ‘to avoid’. If no target whatsoever is in view, the possible actions are ‘to rest’ or ‘to wander’ (see Table I). Affordances are hardwired not learnt. They represent the limitations imposed by the sensory-motor capacities of the model organism, and depend directly on the environment, as understood in the definition of the term (see [3]).

C. Working memory and long-term memory

Working memory (WM) and long-term memory (LTM) are modeled in a highly simplified manner. They nevertheless provide a basic mechanism for associative learning through operant conditioning, implemented via reward and punishment ([32]). WM registers the action selected most recently, and so is updated at each new choice. LTM, by contrast, records each of the organism’s experiences, along with its emotional value, the latter attributed by Full Appraisal (see Figure 1), following the emotional evaluation performed by Hedonic Valuation (see Figure 1). Shortly after an action is taken, its emotional value is determined, given by the difference in deprivation levels of the relevant bodily need immediately prior to and after the action. Actions with a positive value are selected with a probability proportional to this value; those with a negative value are not selected.

D. Action selection

The structure Action Selection selects the next action to be performed depending on the situation, the bodily need to be attended, and the existing affordances. Selection occurs via application of a series of filters, until just one possibility remains. The selection algorithm is summarized below (see Algorithm 1), including the following possibilities: selection based solely on the existing possibilities (affordances); when there are two or more similar targets, only the nearest of each kind is maintained (nearest); selection determined on the basis of the valuation in LTM (memory); finally (if the other criteria do not lead to a unique choice), random selection between alternatives (random).

Algorithm 1 Action Selection Algorithm

Require: Action Tendency and current state of organism
Ensure: Action to be realized

$PossibleActions \leftarrow Affordances(situational)$;
$Drivewithhighestdeprivationlevel \leftarrow verifyGreatest(drives)$;
$Actions \leftarrow PossibleActions \cap ActionTendency(drivewithhighestdeprivationlevel)$;

if $\mid actions \mid > 1$ **then**
  $actions \leftarrow excludeIdenticalTargets(actions)$;
else
  $actionToExecute \leftarrow action$; ▷ (Affordances)
end if
if $\mid actions \mid > 1$ **then**
  if ($\exists$ experience in LTM with object) **then**
    $actionToExecute \leftarrow Random(\text{weighed})LTM(actions)$; ▷ (Memory)
  else
    $actionToExecute \leftarrow Random()actions$; ▷ (Random)
  end if
else
  $actionToExecute \leftarrow actions$; ▷ (Nearest)
end if

IV. SIMULATION

To test the model, we devised an artificial life application in a two-dimensional world using multiple threads to simulate the concurrent behavior of the organism. Dispersed in the environment are three kinds of nutrients (fruits)[11]. We adopt the pixel as our unit of area. The organism, which resembles a PacMan[12], is represented by a blue disk of diameter 30 pixels. Its eye is represented by its visual field in the form of an arc, whose extent varies from 50° to 150°, with an initial value of 70°. The radius of the visual field is fixed at 150 pixels;

[11] The fruits are designated Redfruit, Greenfruit e Grayfruit. Their respective nutritional values are given in Table II.

[12] The well known video game of the 1980s featuring a round head with a mouth that opens and closes. See http://pacman.com/ja/index.html.

TABLE II
VALUES OF THE MAIN PARAMETERS IN SIMULATIONS.

Configuration	Parameter	Value
	Nutritional value	
	Redfruit	0.1
`Final`	Greenfruit	0.2
	Grayfruit	0.0
	Δ	0.005

The nutritional density (fruit /area) was maintained constant via replenishment of eaten fruits.

when the organism is sleeping, the visual field closes. The orientation of the body coincides with the bisector of the visual field. The body is limited to just a head, an eye and a mouth. It has a tactile sense through both its mouth and through the rest of the head. Eating, turning, and moving (with variable speed) are the three actions the organism is capable of. It can also rest, that is, remain in the same place, with its eye shut. Initially the organism is able to perform the following innate actions: wander, that is, move in a direction chosen at random, uniform on the interval between -30° and $+30^\circ$ with respect to its present orientation[13]; avoid, which corresponds to moving in a direction chosen uniformly on the interval between -45° a $+45^\circ$ with respect to its present orientation; approach, that is, move toward an object in the field of vision; rest; eat a fruit that is touching the mouth.

The two-dimensional world inhabited by the organism measures 860 x 720 pixels (width x height), and possesses periodic boundaries. The organism is initially placed at position (430,360); its initial orientation is chosen at random, uniformly over the interval 0° to 360°. Deprivation levels are measured in arbitrary units, such that the maximum level is 7.0; a level above this value results in death of the organism. Initially the deprivation levels are both set to 0.18, because this was considered the minimum level of physiological activity. If the organism does not eat, the number of steps it may take is limited, since its level of hunger increases by a fixed amount, Δ at each step, and it begins with a deprivation level of $P_{min} = 0.18$ and dies when this level reaches $P_{max} = 7$. The number of steps possible without eating is $M_{max} = (P_{max} - P_{min})/\Delta = 1364$ The main screen of the application is shown in Figure 3.

V. COMPUTATIONAL EXPERIMENTS

An initial series of experiments was performed in order to identify a convenient set of parameter values, such that the organism could lives long enough to explore its environment and exhibit learning. Of particular importance are the following parameters: sleeping time of threads; priorities of the threads; nutritive value of the fruits; metabolic rate Δ. The values of these parameters are given in Table II.

[13]At each step, one decision. Avoiding means to turn the body in a random direction (on that interval) and do one step. The same process occurs to the others actions.

The experiments were designed to measure the following properties: lifetime; number of nutrients consumed; total distance traveled; and number of selections realized. Since that all four quantities are strongly correlated, we chose the number of selections, S, as the quantity of reference in analyzing our results for the organism's lifetime. Each experiment consists of fifty independent realizations, allowing us to calculate mean values to reasonable precision. Specifically, the relative uncertainty of the mean (95% confidence interval) is $\frac{E}{\overline{S}} = 0.13$, where the mean number of selections per realization is $\overline{S} = 2.82 \times 10^4$ for the experiment with memory.

In all fifty realizations, 100 fruits (20 redfruits, 20 greenfruits and 60 grayfruits) are distributed in randomly chosen positions. In all studies, fruits are replenished by adding a new fruit of the same kind in a random position each time a fruit is consumed. We verified that the organism explored all the available space, moving in an unbiased manner. The internal aspects of interest in these studies are: the mechanism of action selection; the influence of memory on this mechanism; the temporal evolution of the levels of hunger and need for rest. In the following subsections we discuss our simulation results, seeking the relation between the internal dynamics and behavioral variables (specifically, S).

VI. RESULTS AND DISCUSSION

To understand the role of learning in the behavior of the organism, we performed a set of control or baseline studies in which memory was deactivated, but with all other parameter values (see Table II) as in the studies with memory, referred to hereafter as the *final* experiment. As expected, memory permits a greater level of adaptation (in particular, a longer lifetime).

We observed that the organism develops preferences for different kinds of fruit through its interactions with its environment, and that the evaluation of these experiences was considered necessary and sufficient to establish a minimum criterion of associative learning.

To understand the effect of a nonuniform nutrient density, we performed experiments (with and without memory) with the fruits restricted to half of the environment, i.e., to the left of the midline. All other parameters, including the total number of fruits and the proportions of different kinds of fruits, were maintained at their previous values (i.e., as per Table II). These results indicate that the performance of the organism is poorer with food restricted to half the space, but that memory nevertheless increases the lifetime significantly. (While the mean lifetime in the experiment with a uniform distribution of nutrients with memory was $1.4{\times}10^4 s \quad (\sigma = 7.8 \times 10^3)$ and without memory $2.1{\times}10^3 s \quad (\sigma = 3.1 \times 10^2)$; the mean lifetime in a not uniform environment with memory was $4.1{\times}10^3 s \quad (\sigma = 1.5 \times 10^3)$ and $1.1{\times}10^3 s \quad (\sigma = 1.5 \times 10^2)$ without memory).

The frequencies of the criteria used to select actions are shown in Figure VI.

In Figure VI, illustrating typical realizations from each experiment, the cumulative number of selections is plotted over the lifetime of the organism. Although the total number

TABLE III
RELATIVE UNCERTAINTIES IN NUMBER OF SELECTED ACTIONS, $\overline{S}$ (95% CONFIDENCE INTERVAL).

Property	Action selection		
Experiment	$\overline{S}$	σ	$\frac{E}{\overline{S}}$
`With mem.`	2.8×10^4	1.4×10^4	1.3×10^{-1}
`Without mem.`	0.3×10^4	0.04×10^4	$4 \times 10-2$

σ = standard deviation; $\frac{E}{\overline{S}}$ = standard error relative to mean

of selections differs by approximately 2×10^4, it is evident that both with and without memory, the criteria 'Nearest' and 'Affordances' are used in a very similar manner.

The initial phases of the same experiments are shown in Figure VI, which allow one to follow the evolution prior to the formation of memories. Up to approximately interation 150 of the final experiment, memory is not used to select action, although memories have begun to be formed; the choice is based principally on random selection and affordances. Following the initial phase, memory is used as a criterion, and random choices are no longer needed. In the control group, without memory, the selection criteria do not exhibit any trend over time.

In a typical realization of the final experiment (see Figure 7), the organism regulated its levels of hunger and need for sleep.[14] In this realization the organism maintained behavioral homeostasis at a level compatible with its survival over a period of 4×10^4 selected actions. Note that the number of steps far exceeds 1364, the maximum possible without regulating the level of hunger. In the control group realization, shown in Figure 7, the organism survived for only about 2×10^3 selected actions. (In Figure 7, the hunger levels in both realizations - with and without memory - are shown to facilitate comparison.)

The behavioral efficiency is determined by the level of deprivation via (1); the longest-lived realizations, the efficiency was typically between 70% and 100%. Recall that the behavioral efficiency governs the speed of movement, and therefore the ability of the organism to find food. From time to time the organism rests, staying in a fixed position with its eye closed. In the realization shown in Figure 8 (control group) the organism suffered a higher level of deprivation, reducing its efficiency. The consumption of food during the lifetime of the organism is shown in Figure 9.

The lack of synchronization imposes limitations which, while showing a certain robustness of the model, leads to difficulties in the analysis of results, since the repeated execution of a given internal structure generates multiple internal messages. The multiple messages are interpreted by the recipient structure as a single message, thereby reestablishing stability. Besides that, this lack of synchronization complicates processing of stimuli. Nevertheless, the majority of stimuli (84%), produce only local effects, while only 40% of stimuli produce global effects stemming from self-organization of the dynamical system [15].

[14]The graph of need for sleep is very similar to that of hunger, and so is not shown.

VII. CONCLUSION

Based on our simulation results, we conclude that learning increases the lifetime of the virtual organism. A key point of our model is the concurrent nature of its dynamics, the absence of synchronization points, and the fact that these features do not hinder the emergence of adaptive behavior. Thus despite the difficulty resulting from lack of synchronization, we have shown that it is possible to construct a model of an asynchronous dynamical system which allows the emergence of a behavior consistent with the internal dynamics of the nervous system.

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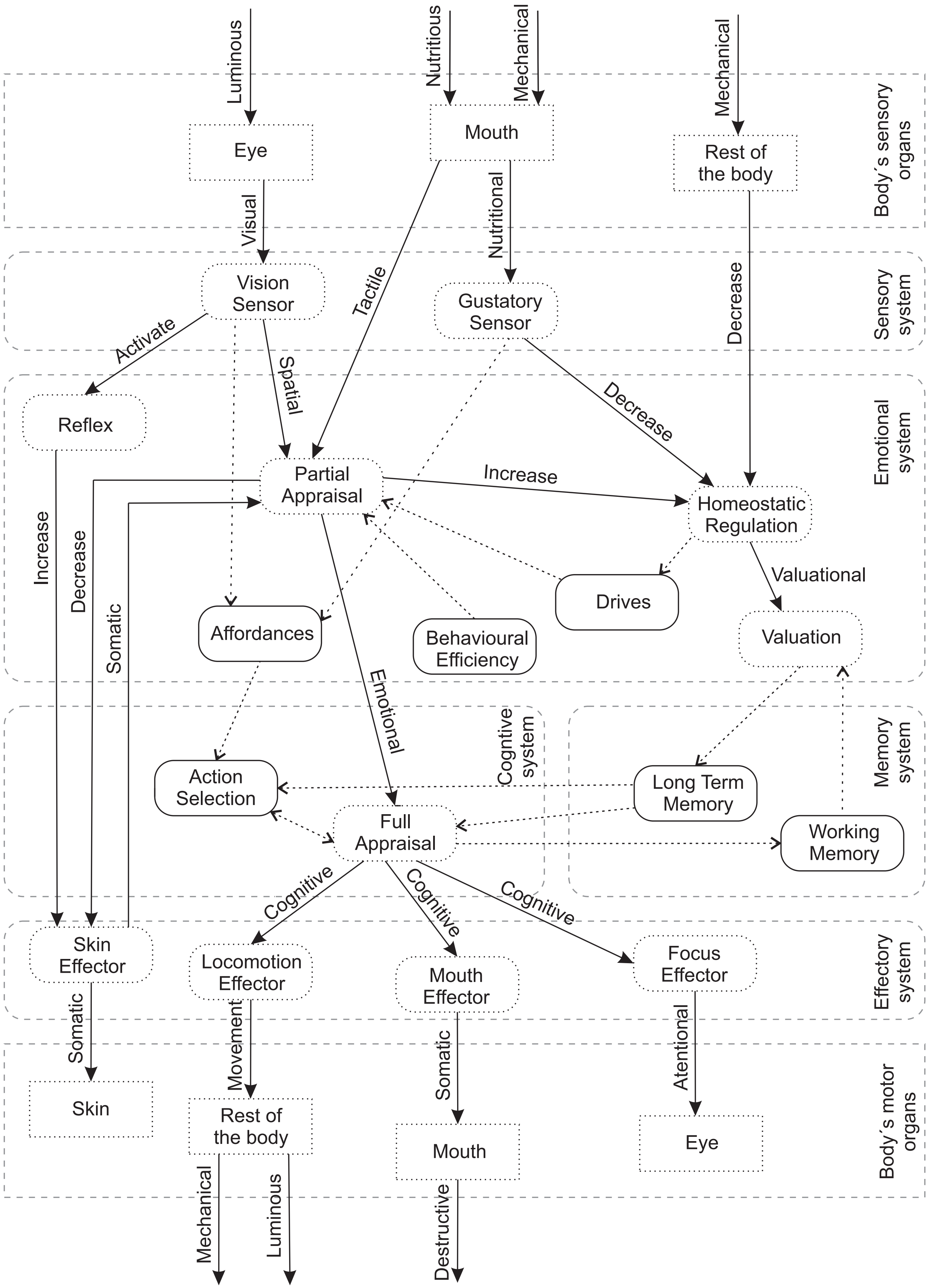

Fig. 1. Stimuli and messages exchanges when a recent action is evaluated.

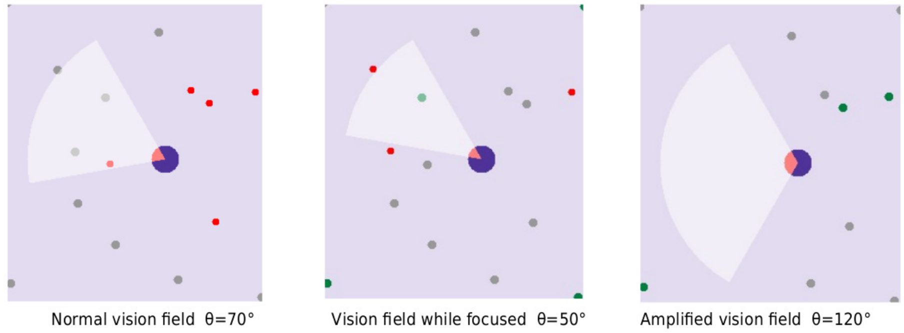

Fig. 2. Variations of the visual field.

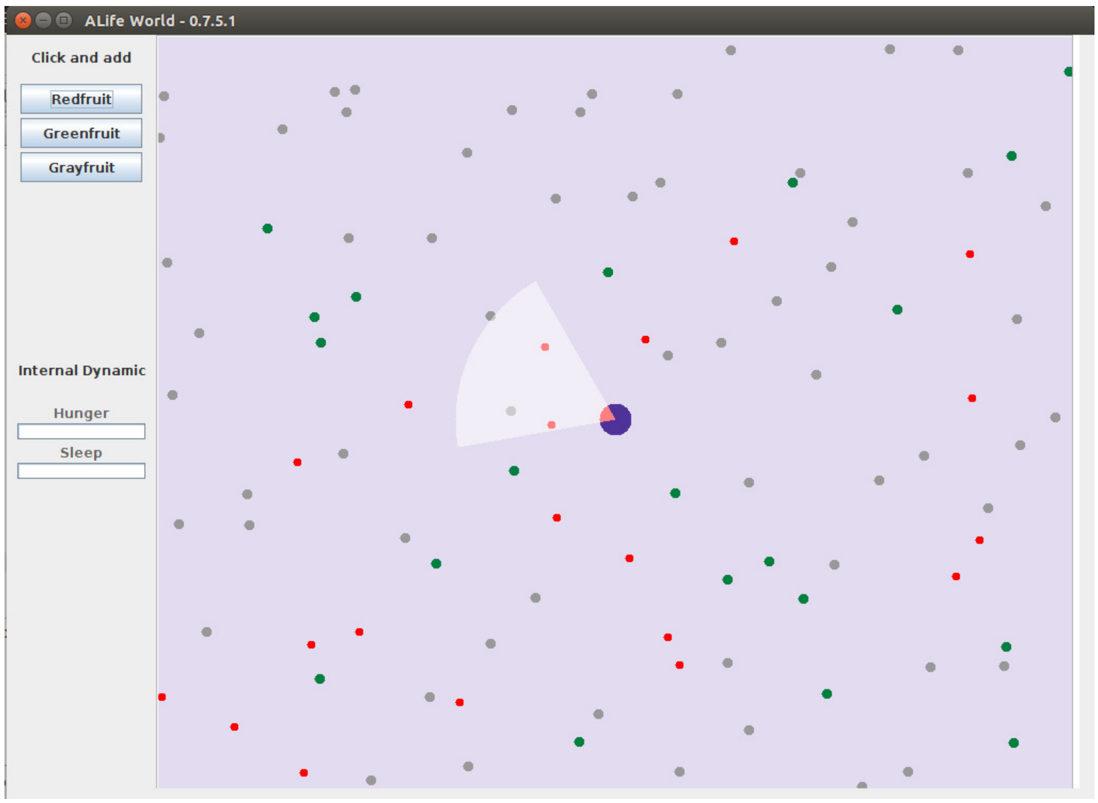

Fig. 3. Main screen of the A-Life application developed for testing the model, with various fruits distributed randomly in space. On the left are bars showing the levels of hunger and need for rest.

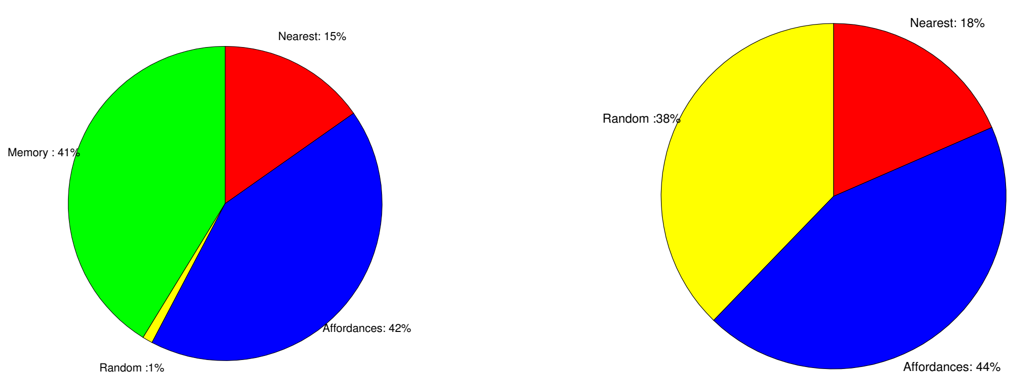

Fig. 4. Frequencies of action selection criteria / Control study in the right side.

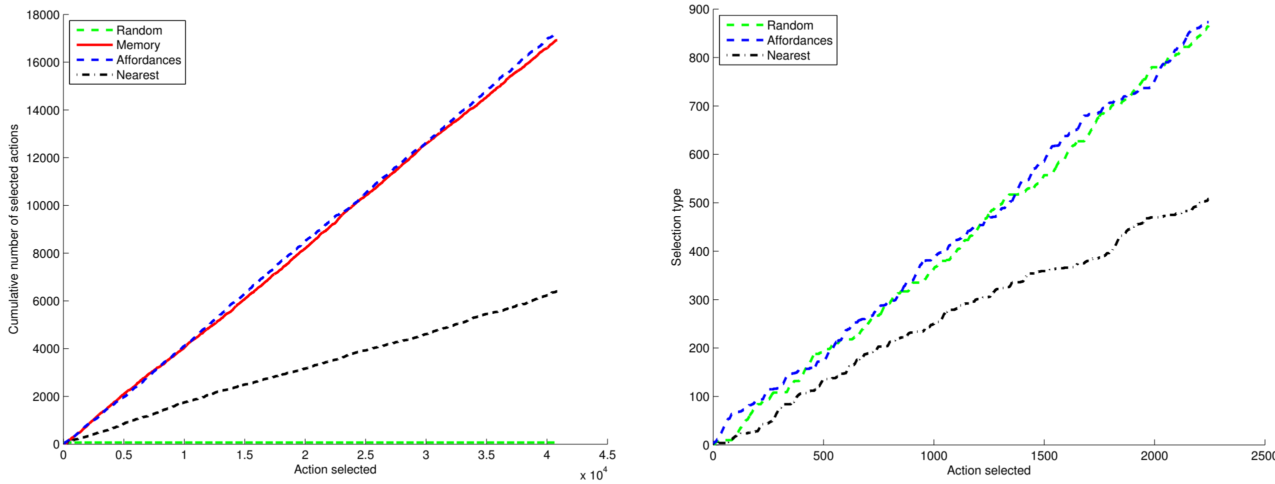

Fig. 5. Cumulative number of actions selected by different criteria / Control study in the right side.

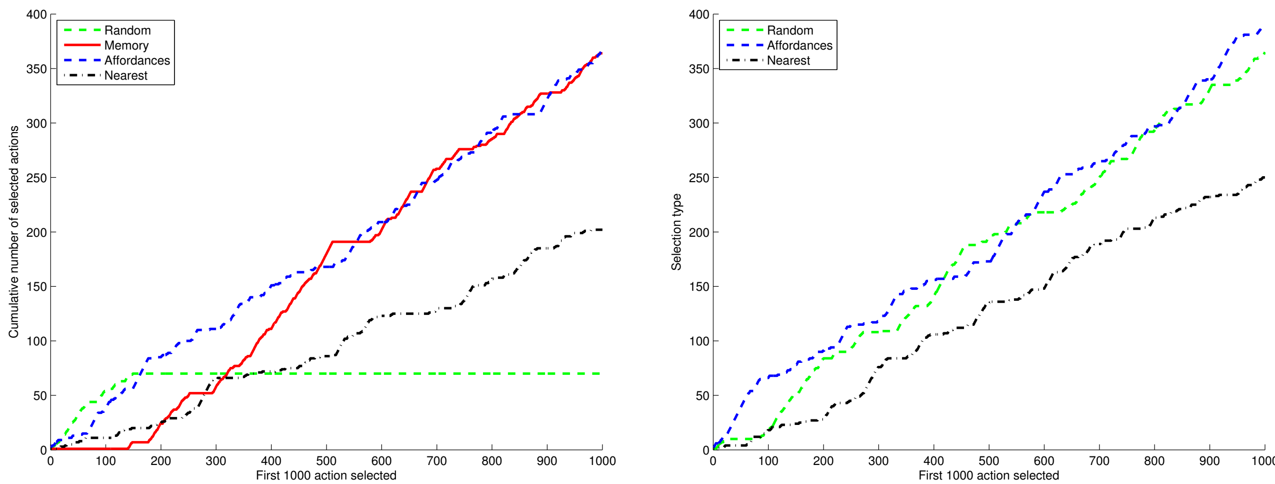

Fig. 6. Cumulative number of actions selected by different criteria (first 1000 interactions) / Control study in the right side.

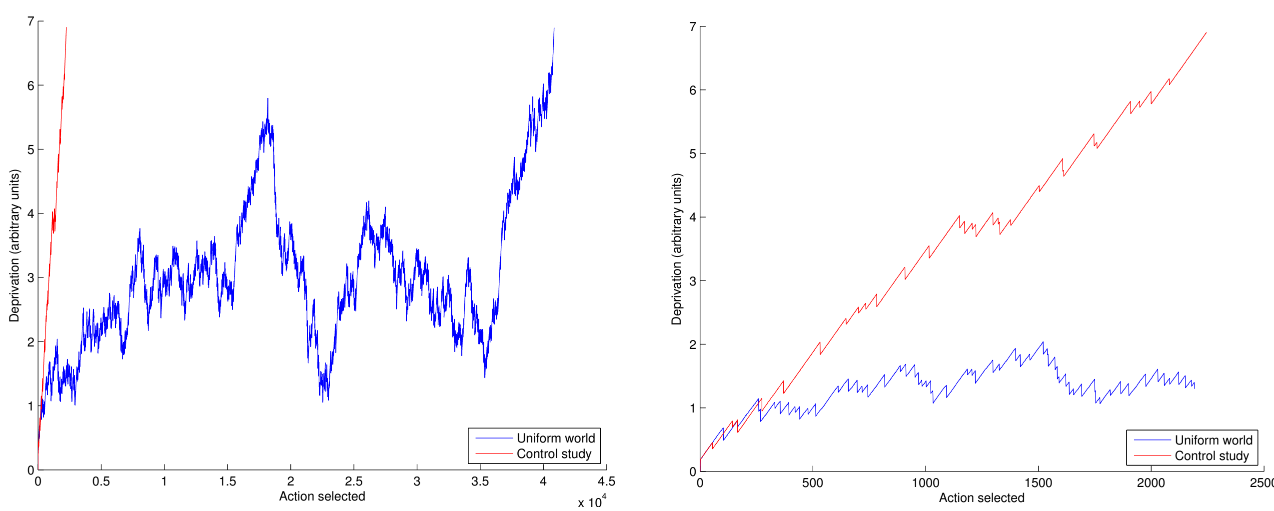

Fig. 7. Deprivation levels versus time / Control study in the right side.

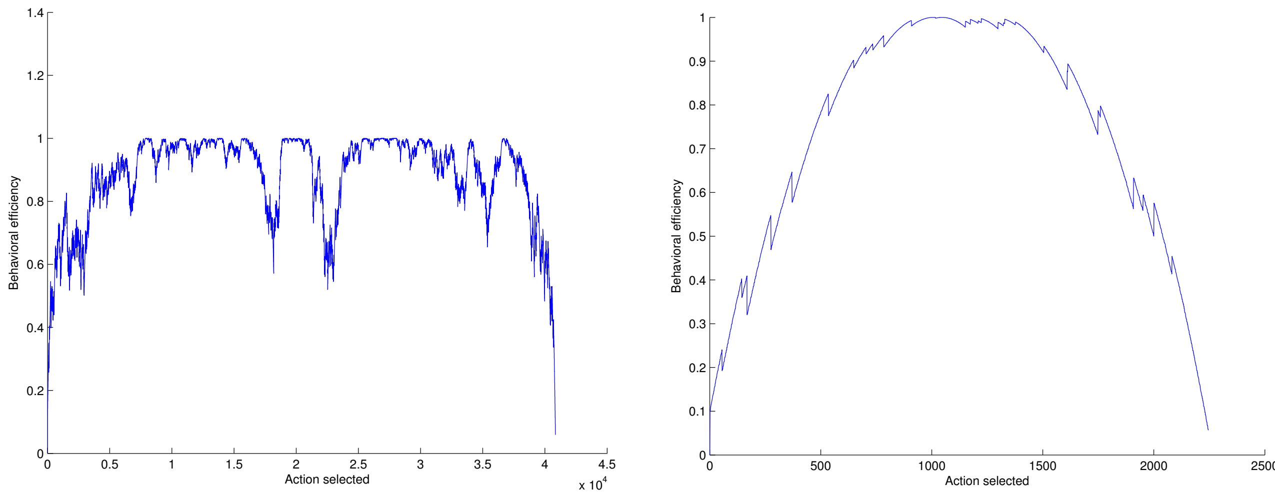

Fig. 8. Behavioral efficiency versus time / Control study in the right side.

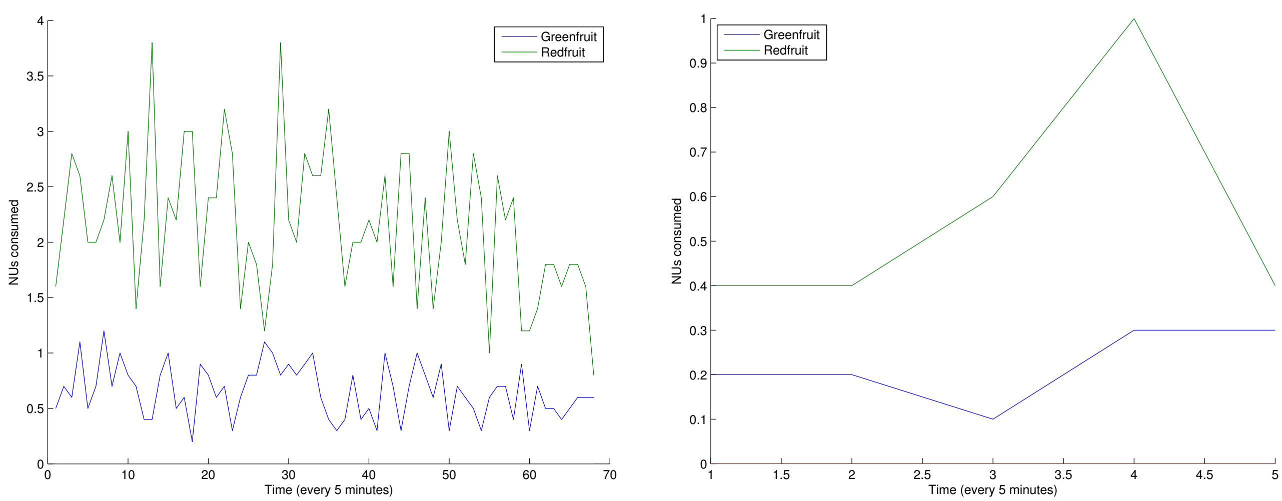

Fig. 9. Nutritional intake versus time / Control study in the right side.